

COMBINATORICS

EXAM 2

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1. (25pts): Let a_n be the number of ways to partition an n -element set into ordered pairs (i.e. matchings where order within each pair matters). Find the exponential generating function for a_n . Derive a closed-form expression for a_n .

Solution. The pairs themselves are unordered, but the two entries in each pair are ordered. On a fixed two-element set there are $2!$ ordered pairs, so the exponential generating function for one pair is $2!x^2/2! = x^2$. The exponential formula therefore gives

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = \exp(x^2) = \sum_{m \geq 0} \frac{x^{2m}}{m!}.$$

Comparing coefficients yields

$$a_{2m} = \frac{(2m)!}{m!}, \quad a_{2m+1} = 0.$$

Here $a_0 = 1$, corresponding to the empty partition. Alternatively, arrange the $2m$ elements in a list and split it into consecutive pairs. Each partition is counted $m!$ times, once for each ordering of its pairs, giving the same formula. $\square$

2. (25pts): Let a_n be the number of ways to distribute n labeled objects into any number of labeled boxes such that each box contains at least two objects. Construct the exponential generating function for a_n . Extract a formula for a_n .

Solution. Interpret “any number of labeled boxes” as summing over $k \geq 0$, with the boxes labeled $1, \dots, k$ for each fixed k . (Choosing arbitrary labels from an infinite supply would not give a finite count.) A single box contains an unordered set of at least two labeled objects, with exponential generating function

$$B(x) = \sum_{j \geq 2} \frac{x^j}{j!} = e^x - 1 - x.$$

For k labeled boxes the generating function is $B(x)^k$. Consequently, as a formal power series,

$$A(x) = \sum_{k \geq 0} (e^x - 1 - x)^k = \frac{1}{2 + x - e^x}.$$

If box i contains n_i objects, the number of distributions is the multinomial coefficient $n!/(n_1! \cdots n_k!)$. Thus an explicit finite formula is

$$a_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \sum_{\substack{n_1 + \cdots + n_k = n \\ n_i \geq 2}} \frac{n!}{n_1! \cdots n_k!}.$$

The $k = 0$ term is 1 when $n = 0$ and 0 otherwise. Equivalently, $a_n = \sum_k k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}_{\geq 2}$, where the symbol counts partitions into k blocks, each of size at least two. $\square$

3. (25pts): The Catalan numbers are defined by $C_n = \frac{1}{1+n} \binom{2n}{n}$. Probe that C_n satisfies the recurrence $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$. Give a combinatorial interpretation of this recurrence.

Proof. Let $C(x) = \sum_{n \geq 0} C_n x^n$. The formal binomial expansion gives

$$\frac{1 - \sqrt{1 - 4x}}{2x} = \sum_{n \geq 0} \left[-\frac{1}{2} \binom{1/2}{n+1} (-4)^{n+1} \right] x^n = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^n.$$

Hence $C(x) = (1 - \sqrt{1 - 4x})/(2x)$, which satisfies $C(x) = 1 + xC(x)^2$. Taking the coefficient of x^n , for $n \geq 1$, proves

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}, \quad C_0 = 1.$$

Combinatorially, C_n counts Dyck paths: paths with n up-steps and n down-steps that never go below height zero. Indeed, the reflection principle counts the paths that go below zero as $\binom{2n}{n+1}$, so the number of Dyck paths is $\binom{2n}{n} - \binom{2n}{n+1} = C_n$. Every nonempty Dyck path decomposes uniquely as $UPDQ$, where the displayed D is the first return to height zero. If P has semilength k , then Q has semilength $n - 1 - k$. There are $C_k C_{n-1-k}$ choices for these two paths; summing over k gives the recurrence. $\square$

4. (25 pts): How many integers between 1 and 1000 (inclusive) are divisible by at least one of 3, 5, or 7? Use the Inclusion-Exclusion Principle and show all steps.

Solution. Let $A_d = \{m \in \{1, \dots, 1000\} : d \mid m\}$. Since 3, 5, 7 are pairwise coprime, their intersections consist of multiples of the corresponding products. Thus

$$\begin{aligned} |A_3| &= 333, & |A_5| &= 200, & |A_7| &= 142, \\ |A_3 \cap A_5| &= \lfloor 1000/15 \rfloor = 66, \\ |A_3 \cap A_7| &= \lfloor 1000/21 \rfloor = 47, \\ |A_5 \cap A_7| &= \lfloor 1000/35 \rfloor = 28, \\ |A_3 \cap A_5 \cap A_7| &= \lfloor 1000/105 \rfloor = 9. \end{aligned}$$

By inclusion–exclusion,

$$\begin{aligned} |A_3 \cup A_5 \cup A_7| &= 333 + 200 + 142 - 66 - 47 - 28 + 9 \\ &= \boxed{543}. \end{aligned}$$

□

5. (25pts): Suppose we have an infinite supply of pennies and we would like to stack them up in rows. Find the number of different arrangements possible subject to the following conditions:

- (i) the bottom row consists of n pennies, each touching its two neighbors, except the ends where $n \geq 1$;
- (ii) a coin that does not belong to a row sits on two coins below it;
- (iii) do not distinguish between heads and tails.

Solution. Interpret condition (ii) as saying that every coin outside the *bottom* row rests on two adjacent coins immediately below it. Upper rows may have gaps, and left–right reflections are counted separately. Let f_n count these stacks on a fixed bottom row of n coins, and set $f_0 = 1$.

There are $n - 1$ possible coin positions immediately above the bottom row. If this entire next row is occupied, the choices above it form a stack with bottom row of length $n - 1$, giving f_{n-1} arrangements.

Otherwise, let k be the first vacant position in that row, counted from the left, where $1 \leq k \leq n - 1$. The first $k - 1$ positions are occupied, and the choices above these coins give f_{k-1} stacks. To the right of the vacancy there is an arbitrary stack on the last $n - k$ bottom coins, giving f_{n-k} choices. No coin can bridge the vacancy: any such coin would require, through its supports, the missing coin. The two choices are therefore independent. Consequently,

$$f_n = f_{n-1} + \sum_{k=1}^{n-1} f_{k-1} f_{n-k} = \sum_{k=1}^n f_{k-1} f_{n-k}.$$

This is the Catalan recurrence with $f_0 = 1$, so

$$f_n = C_n = \frac{1}{n+1} \binom{2n}{n}.$$

For example, $f_1 = 1$, $f_2 = 2$, and $f_3 = 5$. □

6. (25pts): Consider the colorings of the vertices of a regular hexagon using 3 colors. Determine the cycle index polynomial of the dihedral group D_6 . Use Polyá's Enumeration Theorem to compute the number of distinct colorings.

Solution. Write s_j for a cycle of length j . The group D_6 has 12 elements, with the following cycle types on the six vertices:

Symmetry	Number	Cycle monomial
Identity	1	s_1^6
Rotations by $\pm 60^\circ$	2	s_6
Rotations by $\pm 120^\circ$	2	s_3^2
Rotation by 180°	1	s_2^3
Reflections through opposite vertices	3	$s_1^2 s_2^2$
Reflections through opposite edges	3	s_2^3

Averaging these monomials gives

$$Z(D_6) = \frac{1}{12}(s_1^6 + 2s_6 + 2s_3^2 + 4s_2^3 + 3s_1^2 s_2^2).$$

Each cycle must be monochromatic in a coloring fixed by its permutation. With three available colors, Polya's theorem therefore gives

$$Z(D_6)(3, 3, 3, 3, 3, 3) = \frac{3^6 + 2(3) + 2(3^2) + 4(3^3) + 3(3^4)}{12} = \frac{1104}{12} = \boxed{92}.$$

Colors are distinct, and there is no requirement to use all three. $\square$

7. (25pts): Model a benzene ring as a hexagon. Suppose we substitute hydrogen atoms with two types of atoms (A and B). Using Polya's Enumeration Theorem, determine the number of distinct substitution patterns (up to rotational and reflectional symmetry).

Solution. If every site is occupied by either A or B , these are two-colorings of a hexagon under D_6 . Substituting $s_j = 2$ into the cycle index from Problem 6 gives

$$Z(D_6)(2, 2, 2, 2, 2, 2) = \frac{2^6 + 2(2) + 2(2^2) + 4(2^3) + 3(2^4)}{12} = 13.$$

More precisely, substituting $s_j = a^j + b^j$ gives the pattern inventory

$$\begin{aligned} P(a, b) &= \frac{1}{12}((a+b)^6 + 2(a^6 + b^6) + 2(a^3 + b^3)^2 \\ &\quad + 4(a^2 + b^2)^3 + 3(a+b)^2(a^2 + b^2)^2) \\ &= a^6 + a^5b + 3a^4b^2 + 3a^3b^3 + 3a^2b^4 + ab^5 + b^6. \end{aligned}$$

Thus, for example, exactly three A atoms and three B atoms give three patterns. If both types must occur, the total is $13 - 2 = 11$.

If the wording instead allows some hydrogens to remain, each site has three states, H, A, B . The answer is then 92, as in Problem 6, including the unsubstituted ring; requiring at least one substitution gives $92 - 1 = 91$ patterns. $\square$

8. (25pts): Show that: $R(3, 5) > 13$ and $R(3, 5) \leq 14$.

Proof. Recall that $R(3, 5)$ is the least N such that every graph on N vertices contains a triangle or an independent set of size 5.

Lower bound. Let G have vertex set $\mathbb{Z}/13\mathbb{Z}$, with u adjacent to v exactly when $u - v \in \{1, -1, 5, -5\}$ modulo 13. This graph is triangle-free. Indeed, the neighbors of 0 are 1, 5, 8, 12, and the positive differences between distinct members are 3, 4, 7, 11; none is congruent to ± 1 or ± 5 . Thus no two neighbors of 0 are adjacent, and translation proves the same statement at every vertex.

Suppose five vertices were independent. List them in cyclic order, and let $g_1, \dots, g_5$ be their positive cyclic gaps. No gap is 1, and their sum is 13. Thus, up to reordering, the only possibilities are

$$(5, 2, 2, 2, 2), \quad (4, 3, 2, 2, 2), \quad (3, 3, 3, 2, 2).$$

The first possibility contains a gap of 5, which is an edge. In the second, the gap 3 must have a neighboring gap 2, since there is only one gap 4. In the third, a gap 2 and a gap 3 must be consecutive somewhere around the cycle. In either case, two consecutive gaps sum to 5, again giving an edge between two selected vertices. This contradiction proves that G has no independent set of size 5. Coloring its edges red and its nonedges blue gives no red triangle and no blue K_5 , so $R(3, 5) > 13$.

Upper bound. First, every graph on six vertices has a triangle or an independent triple. To see this, color edges red and nonedges blue. A vertex has at least three incident edges of one color. Among their other endpoints, an edge of that color completes a monochromatic triangle; if none exists, the endpoints themselves form a triangle of the other color. Thus $R(3, 3) \leq 6$.

Next, a triangle-free graph on nine vertices must have an independent set of size 4. Otherwise every vertex has degree at most 3, since its neighbors are independent. If a vertex had degree at most 2, it would have at least six other nonneighbors; by the preceding result these contain an independent triple, which together with the vertex gives an independent set of size 4. All degrees would therefore be 3, impossible because their sum would be 27, an odd number. This proves $R(3, 4) \leq 9$.

Finally, let G be triangle-free on 14 vertices, and choose a vertex v . If $\deg(v) \geq 5$, five of its neighbors are an independent set. Otherwise v has at least $13 - 4 = 9$ other nonneighbors. Their induced graph is triangle-free, so it contains an independent set of size 4; adjoining v gives an independent set of size 5. Hence $R(3, 5) \leq 14$, and together the two bounds yield

$$R(3, 5) = 14.$$

□